12TH CHEMISTRY · CHAPTER 33 · FULL CHAPTER MCQS

Water

104 topic-wise MCQs from the complete chapter with answers and reasons

33.1 PROPERTIES AND QUALITY OF WATER

1. The water molecule is a:

(a) Non-polar molecule
(b) Permanent dipole
(c) Linear molecule
(d) Free radical

Answer: (b) Permanent dipole — Its polarity gives water its unique properties.

2. The forces between water molecules that give it a high specific heat capacity are:

(a) Covalent bonds only
(b) Ionic bonds
(c) Metallic bonds
(d) Hydrogen bonds

Answer: (d) Hydrogen bonds — They result from the polarity of the molecule.

3. Hydrogen bonding gives water a high:

(a) Heat of vaporisation
(b) Electrical conductivity
(c) Redox potential
(d) BOD

Answer: (a) Heat of vaporisation — And a high specific heat capacity.

4. Water dissolves ionic substances by:

(a) Reducing the ions
(b) Oxidising the ions
(c) Surrounding the ions
(d) Precipitating the ions

Answer: (c) Surrounding the ions — Polar water molecules hydrate the ions.

5. Which ions are readily dissolved by water?

(a) DDT molecules
(b) Nitrates and phosphates
(c) PCB molecules
(d) Hydrocarbons

Answer: (b) Nitrates and phosphates — They are ionic.

6. The value of K_w at 25 °C is:

(a) 1.0×10^{-10}
(b) 1.0×10^{-7}
(c) 1.0×10^{14}
(d) 1.0×10^{-14}

Answer: (d) 1.0×10^{-14} — Self-ionisation of water.

7. The ability of water to act as both an acid and a base is its:

(a) Amphoteric nature
(b) Thermal stability
(c) Redox potential
(d) Polarity

Answer: (a) Amphoteric nature — It sets the baseline of the pH scale.

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8. Acid rain increases the concentration of which ion in water?

(a) Na^+
(b) OH^-
(c) H_3O^+
(d) Cl^-

Answer: (c) H_3O^+ — Hydronium ions lower the pH.

9. Acidity mobilises toxic metals from sediments by increasing their:

(a) Atomic mass (b) Density
(c) Colour (d) Solubility

Answer: (d) Solubility — Through protonation or displacement from mineral surfaces.

10. DO stands for:

(a) Dissolved oxygen (b) Direct oxidation
(c) Dense organics (d) Dilute oxide

Answer: (a) Dissolved oxygen — The oxygen available for aquatic organisms.

11. High DO levels generally indicate:

(a) Heavy sewage pollution
(b) Clean, unpolluted water
(c) Industrial waste
(d) High organic load

Answer: (b) Clean, unpolluted water — Low DO suggests pollution.

12. BOD is usually measured over:

(a) Ten days at 0 °C
(b) One day at 100 °C
(c) Five days at 20 °C
(d) Five hours at 37 °C

Answer: (c) Five days at 20 °C — The standard period for BOD.

13. BOD measures the oxygen required by:

(a) Microorganisms (b) Fish
(c) Chemical oxidising agents (d) Green plants

Answer: (a) Microorganisms — To break down organic matter.

14. An oxidising agent used in the COD test is:

(a) $CaCO_3$
(b) NaCl
(c) $K_2Cr_2O_7$
(d) NH_3

Answer: (c) $K_2Cr_2O_7$ — $KMnO_4$ is also used.

15. COD is useful for assessing the:

(a) Colour of water
(b) Total pollution load
(c) Temperature of water
(d) Hardness of water

Answer: (b) Total pollution load — It covers organic and inorganic oxidisable substances.

16. The DO of very pure water is about:

(a) 100–400 mg/L (b) 0–2 mg/L
(c) 2–4 mg/L (d) 7–10 mg/L

Answer: (d) 7–10 mg/L — Table 33.1.

17. The BOD of drinking water should be:

(a) 10–30 mg/L
(b) Less than 2 mg/L
(c) 100–400 mg/L
(d) More than 500 mg/L

Answer: (b) Less than 2 mg/L — Table 33.1.

18. The BOD of raw sewage is about:

(a) 2–8 mg/L (b) Less than 1 mg/L
(c) 100–400 mg/L (d) 10–30 mg/L

Answer: (c) 100–400 mg/L — Table 33.1.

19. Which water has the highest COD?

(a) Industrial waste (b) Drinking water
(c) Treated sewage (d) Moderately polluted water

Answer: (a) Industrial waste — 1000–20,000+ mg/L.

20. In well-oxygenated water sulphur exists as:

(a) Elemental sulphur
(b) H_2S
(c) S^{2-}
(d) SO_4^{2-}

Answer: (d) SO_4^{2-} — The most oxidised form.

21. In stagnant, oxygen-depleted water the toxic gases formed are:

(a) H_2S and NH_3
(b) O_2 and O_3
(c) CO_2 and N_2
(d) Cl_2 and HCl

Answer: (a) H_2S and NH_3 — The environment becomes reducing.

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33.2 SOURCES OF WATER POLLUTION

22. Pollution entering from a single, identifiable location is:

(a) Thermal pollution
(b) Non-point source pollution
(c) Diffused pollution
(d) Point source pollution

Answer: (d) Point source pollution — For example a wastewater pipe.

23. At a point source, the concentration of pollutant is highest:

(a) Far upstream
(b) At the site of discharge
(c) In the open sea
(d) In the clouds

Answer: (b) At the site of discharge — This gives localised acute toxicity.

24. A leaking storage tank at a chemical plant is a:

(a) Point source (b) Non-point source
(c) Diffused source (d) Weather source

Answer: (a) Point source — It is one identifiable location.

25. Non-point source pollution is also called:

(a) Pipe pollution (b) Plume pollution
(c) Diffused pollution (d) Acute pollution

Answer: (c) Diffused pollution — It comes from a wide area.

26. Non-point source pollution depends heavily on:

(a) Tank size (b) Factory timings
(c) Pipe diameter (d) Weather events

Answer: (d) Weather events — Rainfall or snowmelt carries the pollutants.

27. Compared with a point source, NPS pollution gives:

(a) Lower concentrations over a wider area
(b) Higher concentrations at one point
(c) A single pollution plume
(d) No pollutants at all

Answer: (a) Lower concentrations over a wider area — A mixture of pollutants is spread out.

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28. Which is a non-point source?

(a) A factory wastewater pipe
(b) Run-off from farmland
(c) A sewage outfall pipe
(d) A leaking storage tank

Answer: (b) Run-off from farmland — It comes from thousands of hectares.

29. Stormwater from roads carries zinc from:

(a) Fertilizers (b) Paints
(c) Tyres (d) Detergents

Answer: (c) Tyres — It also carries engine oils and road salts.

30. The fertilizer named as washed from farmland by rain is:

(a) $Al_2(SO_4)_3$
(b) NaCl
(c) $CaSO_4$
(d) NH_4NO_3

Answer: (d) NH_4NO_3 — Ammonium nitrate.

33.3 COMMON WATER POLLUTANTS

31. Which is a pathogen found in polluted water?

(a) Spirogyra (b) Penicillium
(c) Yeast (d) Escherichia coli

Answer: (d) Escherichia coli — Also Salmonella and Giardia.

32. Which is a water-borne disease?

(a) Diabetes (b) Malaria
(c) Cholera (d) Anaemia

Answer: (c) Cholera — Also typhoid, dysentery and hepatitis A.

33. Boiling water removes:

(a) Most bacteria, viruses and protozoa
(b) Heavy metals
(c) Nitrates
(d) Pesticides

Answer: (a) Most bacteria, viruses and protozoa — Other pollutants are not removed by boiling.

34. Which is a heavy metal pollutant?

(a) Sodium (b) Arsenic
(c) Calcium (d) Potassium

Answer: (b) Arsenic — With mercury, lead and cadmium.

35. A major source of mercury in water is:

(a) Lead pipes (b) Metal plating
(c) Coal combustion (d) Phosphate fertilizers

Answer: (c) Coal combustion — Also gold mining and battery manufacture.

36. In sediments, bacteria convert inorganic mercury into:

(a) Mercury metal (b) Methylmercury
(c) Mercury oxide (d) Mercury chloride

Answer: (b) Methylmercury — It is more dangerous.

37. Lead ions often enter water through:

(a) Snowmelt (b) Algal blooms
(c) Oil spills (d) Corrosion of pipes

Answer: (d) Corrosion of pipes — Old plumbing used lead pipes.

38. Which is a source of cadmium?

(a) Metal plating (b) Leaded petrol
(c) Thermometers (d) Gold mining

Answer: (a) Metal plating — Also plastic stabilisers and phosphate fertilizers.

39. Heavy metals are dangerous mainly because they are:

(a) Volatile
(b) Easily biodegraded
(c) Non-biodegradable and accumulate in tissues
(d) Colourless

Answer: (c) Non-biodegradable and accumulate in tissues — They build up in the food chain.

40. Minamata disease is caused by:

(a) Cadmium (b) Methylmercury
(c) Lead (d) Nitrates

Answer: (b) Methylmercury — A potent neurotoxin.

41. Lead causes anaemia because it interferes with the production of:

(a) Bile (b) Insulin
(c) Calcium (d) Haemoglobin

Answer: (d) Haemoglobin — It is also a cumulative poison in children.

42. Cadmium damages mainly the bones and the:

(a) Kidneys (b) Eyes
(c) Lungs (d) Teeth

Answer: (a) Kidneys — Nephrotoxicity.

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43. The Minamata disaster occurred in:

(a) Pakistan (b) China
(c) India (d) Japan

Answer: (d) Japan — In the 1950s.

44. The WHO limit of arsenic in drinking water is:

(a) 0.05 mg/L (b) 0.01 mg/L
(c) 2.0 mg/L (d) 0.003 mg/L

Answer: (b) 0.01 mg/L — Table 33.2.

45. The WHO limit of copper in drinking water is:

(a) 0.006 mg/L (b) 0.01 mg/L
(c) 2.0 mg/L (d) 0.05 mg/L

Answer: (c) 2.0 mg/L — Table 33.2.

46. Which metal has the lowest WHO limit in Table 33.2?

(a) Cadmium (b) Copper
(c) Chromium (d) Lead

Answer: (a) Cadmium — 0.003 mg/L.

47. The WHO limit of mercury in drinking water is:

(a) 0.06 mg/L (b) 0.006 mg/L
(c) 0.6 mg/L (d) 2.0 mg/L

Answer: (b) 0.006 mg/L — Table 33.2.

48. DDT is a:

(a) Pesticide (b) Heavy metal
(c) Fertilizer (d) Coagulant

Answer: (a) Pesticide — A persistent organic pollutant.

49. POPs stands for:

(a) Point of pollution sources
(b) Primary oxygen pollutants
(c) Polar organic particles
(d) Persistent organic pollutants

Answer: (d) Persistent organic pollutants — Many older pesticides like DDT.

50. DDT is highly soluble in:

(a) Dilute acids (b) Water
(c) Fatty tissues (d) Salt solution

Answer: (c) Fatty tissues — It is non-polar.

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51. Chemicals that behave like natural hormones are called:

(a) Disinfectants (b) Coagulants
(c) Flocculants (d) Endocrine disruptors

Answer: (d) Endocrine disruptors — Many pesticides act like estrogen.

52. Thinning of eggshells in birds is caused by:

(a) Nitrates (b) Pesticides
(c) Oil (d) Microplastics

Answer: (b) Pesticides — Reproductive failure in wildlife.

53. PCBs were used in electrical transformers as:

(a) Coolants (b) Fuels
(c) Conductors (d) Catalysts

Answer: (a) Coolants — They leak from hazardous waste sites.

54. A skin condition caused by PCBs is:

(a) Ataxia (b) Xanthopsia
(c) Chloracne (d) Anaemia

Answer: (c) Chloracne — They are also probable carcinogens.

55. Eutrophication is caused mainly by excess:

(a) Sodium and chlorine (b) Nitrogen and phosphorus
(c) Lead and mercury (d) Oil and plastics

Answer: (b) Nitrogen and phosphorus — From run-off and wastewater.

56. The rapid growth of algae in nutrient-rich water is called:

(a) Algal bloom (b) Dead zone
(c) Pollution plume (d) Floc

Answer: (a) Algal bloom — It blocks sunlight.

57. Hypoxic regions of water are called:

(a) Wetlands (b) Plumes
(c) Reservoirs (d) Dead zones

Answer: (d) Dead zones — Aquatic life cannot survive there.

58. Blue Baby Syndrome is also called:

(a) Minamata disease (b) Nephrotoxicity
(c) Methaemoglobinaemia (d) Itai-itai disease

Answer: (c) Methaemoglobinaemia — Caused by nitrates in drinking water.

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59. An oil layer on the sea:

(a) Blocks sunlight and reduces oxygen exchange
(b) Increases dissolved oxygen
(c) Feeds the fish
(d) Softens the water

Answer: (a) Blocks sunlight and reduces oxygen exchange — It harms marine life.

60. Plastics in the ocean:

(a) Biodegrade within days
(b) Break into smaller particles that persist
(c) Dissolve completely
(d) Evaporate

Answer: (b) Break into smaller particles that persist — They last for centuries.

33.4 POLLUTION DUE TO HUMAN ACTIVITIES

61. The major source of water pollution is:

(a) Recreational activities (b) Industrial activities
(c) Rainfall (d) Fishing

Answer: (b) Industrial activities — Stated in section 33.4.

62. Which industry is named as releasing water pollution?

(a) Tanneries (b) Bakeries
(c) Tailoring (d) Printing of books

Answer: (a) Tanneries — Also refineries, chemical plants and fertilizer plants.

63. Poor irrigation practices increase:

(a) Water purity (b) Dissolved oxygen
(c) Rainfall (d) Soil erosion

Answer: (d) Soil erosion — Sediments are washed into rivers.

64. Untreated sewage introduces into water:

(a) Noble gases
(b) Dissolved oxygen
(c) Harmful bacteria and viruses
(d) Distilled water

Answer: (c) Harmful bacteria and viruses — Domestic and municipal waste.

65. Deforestation pollutes water by increasing:

(a) Fish population (b) Tree cover
(c) Rain purity (d) Sediment in rivers

Answer: (d) Sediment in rivers — Loose soil is eroded.

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66. Boats pollute water by releasing:

(a) Oil and fuel (b) Oxygen
(c) Chlorine (d) Fertilizers

Answer: (a) Oil and fuel — Recreational activities.

67. Hull cleaning and painting of ships release:

(a) Nutrients only (b) Toxic substances
(c) Pure water (d) Oxygen

Answer: (b) Toxic substances — Marine transport.

33.5 EFFECT ON THE ENVIRONMENT

68. A low oxygen level in water is called:

(a) Hypoxia (b) Ataxia
(c) Osmosis (d) Salinity

Answer: (a) Hypoxia — It makes survival difficult for aquatic life.

69. The build-up of toxins in an organism over time is:

(a) Sedimentation (b) Biodegradation
(c) Bioaccumulation (d) Coagulation

Answer: (c) Bioaccumulation — For example mercury and pesticides.

70. The increase of toxin concentration up the food chain is:

(a) Disinfection (b) Eutrophication
(c) Flocculation (d) Biomagnification

Answer: (d) Biomagnification — Top predators get the highest dose.

71. The highest doses of biomagnified toxins are received by:

(a) Algae (b) Top predators
(c) Small fish (d) Plankton

Answer: (b) Top predators — Including humans.

72. Plants given polluted water may show:

(a) Larger fruit
(b) Faster growth
(c) More flowers
(d) Stunted growth and leaf discoloration

Answer: (d) Stunted growth and leaf discoloration — Soil fertility is damaged.

73. The gas released when organic matter decays in anaerobic water is:

(a) Chlorine (b) Oxygen
(c) Methane (d) Ozone

Answer: (c) Methane — A potent greenhouse gas.

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74. Methane emission from polluted water contributes to:

(a) Global warming (b) Ozone formation
(c) Water hardness (d) Acid rain

Answer: (a) Global warming — It is a greenhouse gas.

33.6 WASTEWATER TREATMENT

75. Primary treatment of wastewater removes:

(a) Viruses only (b) Dissolved ions
(c) Physical debris (d) Hardness

Answer: (c) Physical debris — The first level of treatment.

76. Secondary treatment is a:

(a) Biological processing (b) Chemical purification
(c) Screening (d) Reverse osmosis

Answer: (a) Biological processing — The second level.

77. Tertiary treatment is a:

(a) Biological processing (b) Chemical purification
(c) Removal of debris (d) Screening

Answer: (b) Chemical purification — The third level.

78. Large insoluble particles are removed by passing water through:

(a) Activated carbon (b) Ion-exchange columns
(c) Semi-permeable membranes (d) Metal screens

Answer: (d) Metal screens — Screening.

79. In sedimentation, suspended solids settle under:

(a) Electric current (b) Pressure
(c) Gravity (d) Magnetism

Answer: (c) Gravity — Water is held in large tanks.

80. The masses formed by coagulation are called:

(a) Floc (b) Sludge gas
(c) Plume (d) Bloom

Answer: (a) Floc — Small particles clump together.

81. Coagulation is used to speed up the settling of:

(a) Large stones
(b) Fine colloidal particles
(c) Dissolved gases
(d) Bacteria only

Answer: (b) Fine colloidal particles — It also precipitates heavy metals and phosphates.

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82. In filtration, water passes through layers of:

(a) Iron and zinc
(b) Salt and sugar
(c) Clay and lime
(d) Sand and gravel

Answer: (d) Sand and gravel — The sand traps particles.

83. For advanced filtration the material used is:

(a) Gypsum
(b) Rock salt
(c) Activated carbon
(d) Limestone

Answer: (c) Activated carbon — Stated in section 33.6.

84. The most common method of disinfection is:

(a) Sedimentation
(b) Boiling
(c) Filtration
(d) Chlorination

Answer: (d) Chlorination — Ozone and NaOCl are also used.

85. The species that kills bacteria in chlorinated water is:

(a) HOCl
(b) NaCl
(c) HCl gas
(d) ClO_4^-

Answer: (a) HOCl — Hypochlorous acid penetrates cell walls.

86. Which is NOT used as a disinfectant?

(a) Ozone
(b) Aluminium sulphate
(c) Sodium hypochlorite
(d) Chlorine

Answer: (b) Aluminium sulphate — It is a coagulant.

87. Reverse osmosis uses a:

(a) Semi-permeable membrane
(b) Sand bed
(c) Metal screen
(d) Coagulant

Answer: (a) Semi-permeable membrane — Water is forced through under pressure.

88. Reverse osmosis is suitable for regions with high:

(a) Altitude
(b) Rainfall
(c) Salinity
(d) Forest cover

Answer: (c) Salinity — It removes dissolved salts.

89. Hard water contains high concentrations of:

(a) H^+ and OH^-
(b) Na^+ and K^+
(c) Cl^- and F^-
(d) Ca^{2+} and Mg^{2+}

Answer: (d) Ca^{2+} and Mg^{2+} — They interfere with soap.

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90. Softening of water is done by:

(a) Screening
(b) Ion exchange
(c) Chlorination
(d) Sedimentation

Answer: (b) Ion exchange — Hardness ions are swapped for Na^+ or H^+.

33.7–33.8 CONSERVATION AND LAWS

91. STP stands for:

(a) Solid treatment process
(b) Standard temperature plant
(c) Sewage treatment plant
(d) Sludge transfer pipe

Answer: (c) Sewage treatment plant — STPs are upgraded for pollution control.

92. ETP stands for:

(a) Energy transfer plant
(b) Effluent treatment plant
(c) Emission testing point
(d) Eutrophication treatment process

Answer: (b) Effluent treatment plant — Installed by industries.

93. A water-saving irrigation method is:

(a) Hosepipe watering
(b) Flood irrigation
(c) Canal overflow
(d) Drip irrigation

Answer: (d) Drip irrigation — Sprinkler irrigation is another.

94. Greywater is water from:

(a) Laundry, dishwashing and bathing
(b) Toilets
(c) Factories
(d) Rivers

Answer: (a) Laundry, dishwashing and bathing — It can be reused.

95. Greywater can be reused for:

(a) Injections
(b) Drinking
(c) Cooking
(d) Toilet flushing

Answer: (d) Toilet flushing — Also landscaping and industrial cooling.

96. Capturing and storing rain for later use is:

(a) Rainwater harvesting
(b) Reverse osmosis
(c) Eutrophication
(d) Chlorination

Answer: (a) Rainwater harvesting — A sustainable method.

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97. Recharge wells help to:

(a) Kill bacteria (b) Soften water
(c) Replenish groundwater (d) Remove oil

Answer: (c) Replenish groundwater — With small dams.

98. The Punjab Environmental Protection Act was made in:

(a) 2019 (b) 1997
(c) 2023 (d) 2025

Answer: (b) 1997 — Table 33.3.

99. The maximum fine under the Punjab Environmental Protection Act, 1997 is:

(a) Rs 500,000 (b) Rs 10,000
(c) Rs 100,000 (d) Rs 5 million

Answer: (d) Rs 5 million — Plus daily fines up to Rs 100,000.

100. The Punjab Water Act was made in:

(a) 2019 (b) 1997
(c) 2023 (d) 2025

Answer: (a) 2019 — Table 33.3.

101. Under the Punjab Water Act, 2019 the imprisonment is up to:

(a) 3 days (b) 3 months
(c) 3 years (d) 13 years

Answer: (b) 3 months — And/or a Rs 500,000 fine.

102. Discharge permits are regulated by the:

(a) Pakistan Penal Code
(b) Punjab Water Act, 2019
(c) Punjab Irrigation, Drainage and Rivers Act, 2023
(d) Punjab Environmental Protection Act, 1997

Answer: (c) Punjab Irrigation, Drainage and Rivers Act, 2023 — Table 33.3.

103. The fine for washing a car with a hosepipe is:

(a) Rs 1,000 (b) Rs 10,000
(c) Rs 100,000 (d) Rs 5 million

Answer: (b) Rs 10,000 — Did You Know box.

104. EPA stands for:

(a) Environmental Protection Agency
(b) Energy Production Authority
(c) Effluent Processing Act
(d) Environmental Pollution Act

Answer: (a) Environmental Protection Agency — EPA Punjab issues the rules.

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